



### Oldest mammal fossils found

Ancient teeth found in England are some of the oldest known fossils from mammals that gave rise to humans. <http://digit.in/Ancestr>



### NASA's Deep Space Gateway power contract

NASA has awarded 5 contracts to start working out how to power the Deep Space Gateway, it's planned outpost at the moon for launches to other worlds <http://digit.in/DSGPower>

immediate change in the elastic stress of the structure, causing seismicity the first time it is filled, or the water seeps into sub-surface levels and over time, due to decreasing pressure, an earthquake occurs.

One of the most significant earthquakes induced by an artificial water body was right here in India. In 1967, the 6.3 magnitude earthquake in Koynanagar is believed to have been caused by the nearby Koyna Dam reservoir. The epicenter, along with the foreshocks and aftershocks were all located near the dam, although the effect of the quake was felt even 230kms away in Mumbai (then Bombay). One of the most significant earthquake disasters, the 2008 Sichuan earthquake that caused approx 68000 deaths, was also believed to have been caused due to similar seismic activity generated from the nearby Zipingpu Dam.

## ENHANCED GEOTHERMAL SYSTEMS

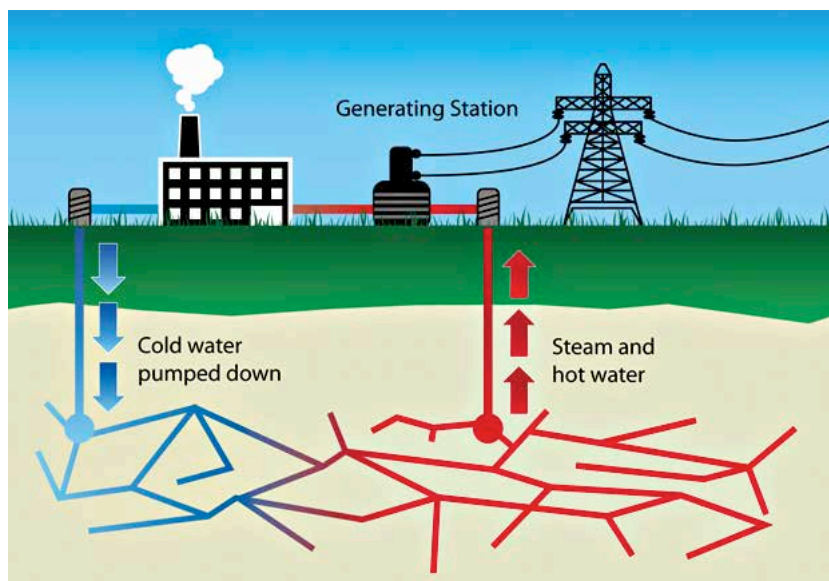
Geothermal energy, or geothermal power, is one of the renewable sources of energy utilised around the world for electricity generation. The way it works is – normal water is passed into the ground, the higher temperature found deep inside the crust heats up that water which is passed back up as steam, which in turn turns a turbine. An improved version of the same process involves pumping pressurised water to enhance permeability, which improves the heating efficiency. This process is known as Enhanced Geothermal System, or EGS.

But due to the rock fracturing and shearing that happens, the rock may respond with structural failure. This

could, and does, lead to some very serious seismic activity around EGS sites. Cerro Prieto, Baja California, Mexico saw activity that went up to 6.6 on the Richter scale. Similar instances elsewhere have even led to cancellation of proposed EGS setups.

## MINING

Mining is a human activity that usually involves digging deep and wide into the Earth's crust. It forms one of the most primary ways of obtaining resources like coal, and several metallic ores. But it also leads to another



How geothermal energy works

impact – earthquakes. Mines generally operate for long period of times, sometimes even going as long as 200 years, as was the case with mines in the New South Wales region. A 5.6 magnitude earthquake struck the region in December 1989, causing more economic damage than the value of the coal that was extracted from the mines.

One of the most significant reasons behind mining-caused earthquakes is the pumping of groundwater to keep the mines from flooding. Volume-wise, groundwater almost four times the volume of coal is pumped out at the low-end of mining process. This affects the stress at the underground level, causing tectonic activity leading to earthquakes.

## WHAT CAN WE DO ABOUT IT?

If we're causing all these earthquakes through so many different activities, there has to be something we can do about it, right? Yes, there are some clearly defined ways in which we can avoid the seismic activity caused by human activity, and they are as follows:

- **Groundwater injection**
  - Setting up monitoring stations
  - Ensuring that such areas are away from high-population zones.
  - Not going after oil/gas extrac-

tion and moving to renewable sources of energy.

- **Artificial Lakes**
  - Not build dams and reservoirs taller than sustainable
- **Mining**
  - Reduced intensity and longevity of mining projects

The thing is, all this has been known to us for ages – for at least as long as we've been causing these earthquakes. But we aren't really excelling at acting on these measures. But there's still time, and we sure do hope that the authorities and governing bodies around the world can come to some agreement about man-made earthquakes. After all, it is about the ground under our very feet slipping away. 



The Koyna Dam